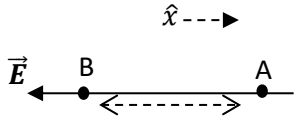
Worksheet # 3 Part B Upload solutions
Calculus Physics II

Name: _____ Date: _____

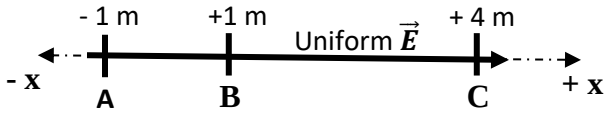
Show all work, no work no credit

<u>Upload the solution to questions 11 through 15 below</u>

11) An object with a net negative charge of 200 μC moves in an electric field $\vec{E}$ that changes along the x axis as $\vec{E} = [(6\,N/Cm^2)x^2]\hat{x}$ from point B at 1 m to point A at 2 m. Determine the work the field does on the charge moving from point B to point A?



12) A uniform electric field is shown below with positions A, B and C.



A) Determine the electric field strength if a particle has a net positive charge of 100 μC and a potential energy difference for the particle between points A and C of $\Delta PE_{C\text{ to }A} = 0.5$ J.

B) Determine the electric potential difference between B and C.

C) If position A is at an electric potential of + 2,000 V, determine the electric potential at point C.

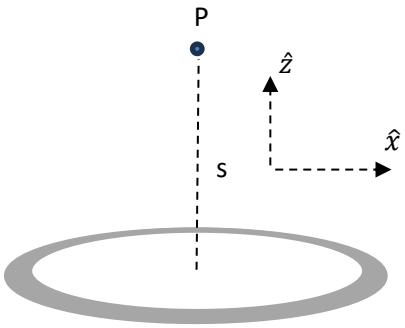
13) The electric potential is changing along the x-axis given by the equation below

$$V(x) = (3\,V/m^3)x^3 - (18\,V/m^2)x^2 + (20\,V/m)x + 2V$$

A) Determine the position(s) along the x axis where the magnitude of the electric field is zero.

B) Determine the potential difference between the zero electric field points.

14) A thin wire of radius R with a uniform linear charge density of λ is shown below, determine the electric potential at point P at a distance of "s" above the center of the wire.



15) An electric motor is connected across an ac voltage source given by the equation below and the rms current that the motor draws is I_{rms} = 4 A.

$$V(t) = (325\ V)\ \sin\ [(377/s)t].$$

A) Determine the resistance of the motor if only the power source and motor are wired in the series.

B) Determine the average power used by the motor.